

■ SMART INDIA HACKATHON 2026 · TECHNICAL DOCUMENTATION REPORT

VARSA

Adaptive Weather Intelligence Engine

"When weather models disagree, VARSA learns which forecast to trust — and how much."

PROBLEM STATEMENT

SIH26081 — Hybrid AI-NWP Multi-Model Forecast
Blending System

TEAM IDENTITY

Arctic Rangers

DOCUMENT CLASSIFICATION

Scientific Architecture & Technical Specification
Report

REPOSITORY IMPLEMENTATION

Deterministic Functional Prototype (v1.0.0)

LIVE PROTOTYPE URL

<https://varsa-demo.vercel.app>

DOCUMENT RELEASE

September 2026 · SIH Evaluation Committee

PROTOTYPE DOCUMENTATION · DEMONSTRATION DATASET

Mandatory Notice: This document details the technical architecture, mathematical blending mechanics, and interface structure implemented in the VARSA repository. The application operates deterministically over a structured demonstration dataset. It does not fabricate live operational capabilities or claim unverified empirical skill improvements over operational meteorological systems.

Multi-Model NWP Foundation: NOAA GFS (0.25°) · ECMWF IFS (0.1°) · DWD ICON (0.125°)

Context Domain: 6 Broad Indian Sub-Divisions · 3 Primary Atmospheric Parameters

Smart India Hackathon 2026

Team Arctic Rangers · Lead Presentation

Annex

REPORT STRUCTURE & NAVIGATION

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DOCUMENT GUIDE FOR TECHNICAL REVIEWERS

This report is specifically formatted as an engineering reference for the Smart India Hackathon 2026 presentation of Problem Statement SIH26081. While the primary PPT presentation focuses on the operational concept and live demonstration, this technical report provides explicit mathematical formulations, component responsibilities, verification metrics definitions, and the scientific roadmap required for operational deployment.

SECTION 02 · STRATEGIC TECHNICAL OVERVIEW

02 Executive Summary

Modern weather forecasting relies fundamentally on Numerical Weather Prediction (NWP) models that simulate fluid atmospheric dynamics through non-linear partial differential equations. Across the Indian subcontinent, operational meteorologists routinely ingest runs from multiple premier international and national forecasting centers—predominantly the NOAA Global Forecast System (GFS), the European Centre for Medium-Range Weather Forecasts Integrated Forecasting System (ECMWF IFS), and the Deutscher Wetterdienst Icosahedral Nonhydrostatic model (DWD ICON).

However, these leading models frequently provide diverging predictions for the exact same geographic sector and issuance window. A heavy rainfall event over the Western Ghats may be intensified by one model due to its convection parameterization, while another suppresses precipitation due to boundary layer friction. Conversely, during pre-monsoon heatwaves across the Indo-Gangetic Plains, thermal advection differences create major temperature discrepancies. In practice, operational forecasters cannot rely on a single global model for every meteorological scenario, nor can they rely on naive equal-weight ensemble averaging, which washes out localized extremes and penalizes the consensus when one model exhibits a known systematic bias.

VARSA (Adaptive Weather Intelligence Engine) is introduced as an adaptive intelligence and blending layer that sits above multiple NWP model inputs. Rather than attempting to replace numerical atmospheric physics with pure statistical regression or black-box neural networks, VARSA accepts harmonized forecasts from multiple dynamical models and dynamically adjusts their relative contributions (w_{GFS} , w_{ECMWF} , w_{ICON}) according to geographic region, forecast lead time, variable physics, and prevailing synoptic weather regimes.

WHAT THE CURRENT PROTOTYPE DEMONSTRATES

- A fully functional, client-side deterministic blending engine implemented in TypeScript.
- Mathematical weighting framework where $\sum w_m = 1.000$ strictly across all dimensions.
- Demonstration scenarios showing dynamic weight adaptation across 4 typical synoptic cases.
- Multi-model spread calculation and categorization (Low / Moderate / High disagreement).
- Verification scorecards (MAE, RMSE, Bias, CSI, POD, FAR) using demonstration reference data.
- Interactive dashboard containing five dedicated operational views for forecaster workflows.

WHAT REMAINS FOR OPERATIONAL VALIDATION

- Ingestion of multi-year historical gridded NWP runs (GFS 0.25°, ECMWF 0.1°, ICON 0.125°).
- Rigorous alignment against IMD gridded rainfall/temperature & automatic weather station (AWS) feeds.
- Training of offline supervised learning models (e.g., gradient boosted regression trees) to derive learned weights.
- Out-of-sample temporal cross-validation preserving strict forecast lead-time causality.
- Operational distributed microservice pipelines for automated GRIB2/NetCDF ingestion.

CLARIFICATION ON PROTOTYPE STATUS

The current VARSA repository is a deterministic functional demonstrator. The numbers, weights, and verification metrics presented in the UI demonstrate the computational architecture and user experience; they do not represent empirically validated forecast skill improvements over operational IMD systems.

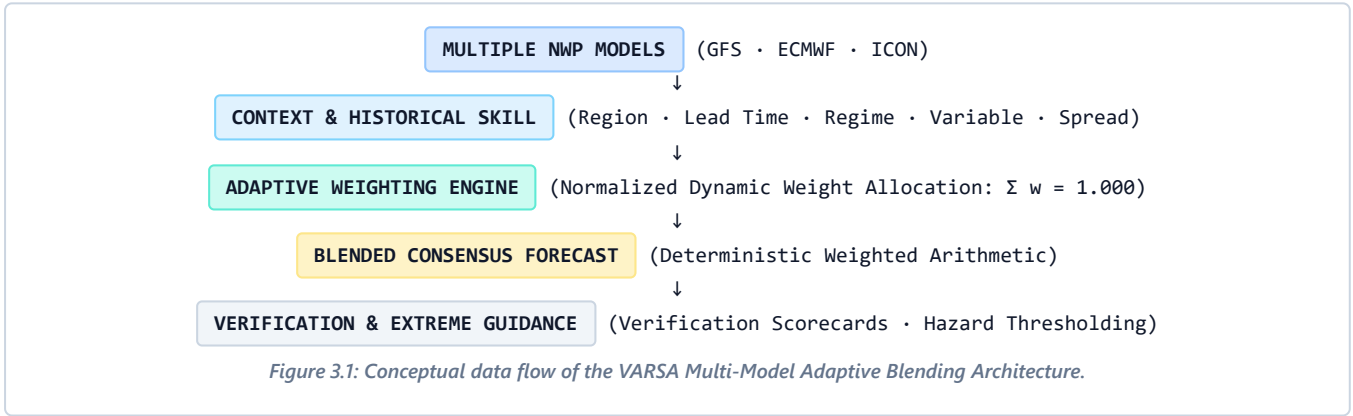
SECTION 03 · METEOROLOGICAL CHALLENGE

03 Problem Definition

Numerical Weather Prediction models differ fundamentally in their dynamical formulations, vertical and horizontal grid resolutions, parameterizations of sub-grid atmospheric physics (such as cloud microphysics, boundary-layer turbulence, and soil-canopy moisture exchange), and observational data assimilation methods. Consequently, model forecast errors are not uniformly distributed in space or time.

Empirical meteorological studies establish that model performance varies systematically across five primary dimensions:

- **Geographic Region & Topography:** Global hydrostatic models struggle with steep orographic barriers (e.g., Western Ghats, Himalayas), whereas non-hydrostatic or higher-resolution mesoscale configurations capture localized convective updrafts more effectively.
- **Forecast Lead Time (T+24h to T+120h):** Mesoscale models frequently outperform in short-range timing (T+0h to T+36h), whereas global spectral models (such as ECMWF IFS) exhibit superior baroclinic stability at extended ranges (T+72h to T+144h).
- **Seasonal & Synoptic Weather Regime:** The dominant atmospheric driver changes error distributions; a model tuned for mid-latitude baroclinic waves may over-predict monsoon depression precipitation cores.
- **Forecast Variable Physics:** A model with superior boundary-layer temperature skill may concurrently display wet biases during convective rainfall events.
- **Ensemble Spread & Disagreement:** High divergence among model runs signals heightened atmospheric unpredictability, demanding conservative or specialized consensus weighting.



Scientific Context: Multi-model super-ensembles and consensus forecasting have an established scientific foundation in meteorology (Krishnamurti et al., 1999; Gneiting & Raftery, 2005). VARSA does not claim that multi-model blending is an unprecedented theoretical concept. Rather, VARSA implements an adaptive framework that dynamically conditions blending weights on operational meteorological context rather than assuming static historical averages.

SECTION 04 · FOUNDATIONAL PARADIGM

04 **VARSA Concept & Foundational Philosophy**

The central design axiom of VARSA is succinct: *“VARSA does not replace weather models — it learns when to trust each one.”*

Numerical weather prediction models solve the Navier-Stokes equations for stratified compressible flow alongside thermodynamic equations. They represent billions of dollars in supercomputing infrastructure, satellite observation networks, and physical parameterization development. Attempting to replace this physics-based numerical foundation with a standalone neural network or statistical model from scratch is both computationally impractical and meteorologically hazardous.

Instead, VARSA operates as an intelligent supervisory layer. It accepts the dynamical projections from established models and focuses entirely on the synthesis problem: determining the optimal linear combination conditioned on the prevailing atmospheric and regional context.

TRADITIONAL OPERATIONAL APPROACH

INDIVIDUAL NWP MODEL → FORECAST ISSUANCE
or
EQUAL WEIGHT ENSEMBLE: (GFS + ECMWF + ICON) / 3

Drawbacks: Disregards localized model biases; dilutes localized convective signals; equal weighting penalizes skill when one model diverges egregiously.

VARSA ADAPTIVE ARCHITECTURE

MULTIPLE MODELS → CONTEXT ANALYSIS
→ ADAPTIVE WEIGHTING → BLENDED FORECAST
→ VERIFICATION & FEEDBACK

Advantage: Exploits complementary model physics; preserves orographic and convective detail; adjusts dynamically to forecast horizon degradation.

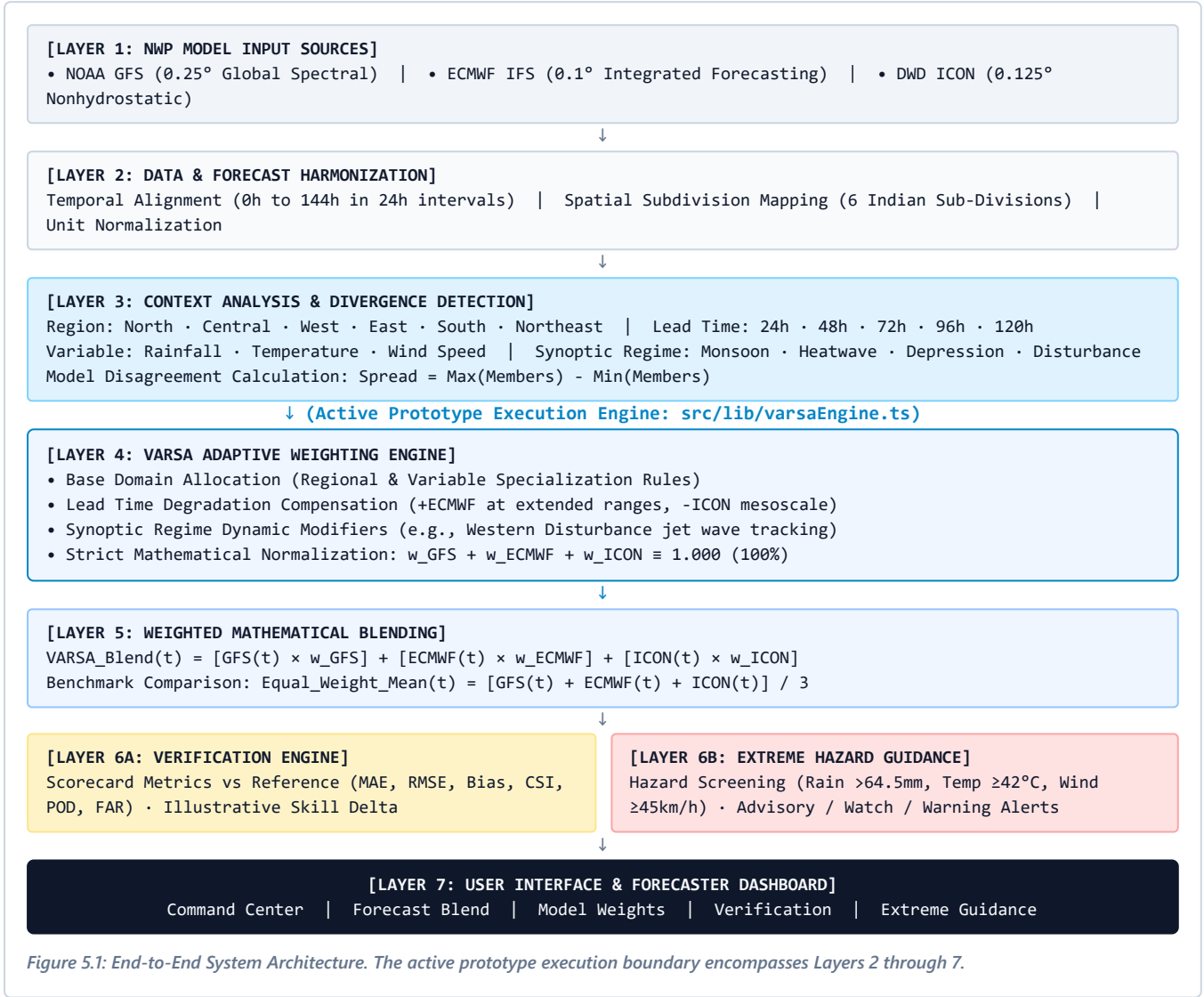
Deterministic Functional Demonstration

The current repository demonstrates this end-to-end concept deterministically. All mathematical equations, normalizations, and weighting calculations execute locally in real time without external API latency. This enables evaluators to inspect the mathematical formulation, test edge cases, and verify that the system behaves coherently across diverse meteorological regimes.

SECTION 05 · PIPELINE & COMPONENT ARCHITECTURE

05 System Architecture

The system architecture is engineered as a decoupled, layered pipeline. Raw numerical weather forecasts flow into a harmonization layer, undergo synoptic context analysis, pass through the adaptive weighting engine, and produce a unified blended forecast paired with verification telemetry and extreme hazard guidance.



Delineation of Prototype Boundary

In the present implementation, Layer 1 uses demonstration member forecasts computed deterministically in `varsaEngine.ts`, representing typical multi-model divergence characteristics across Indian meteorological subdivisions. Layers 2 through 7 are fully implemented in production-grade React and TypeScript code, providing real-time interactive computation, formula verification, responsive thresholding, and vector SVG data visualization.

SECTION 06 · SOURCE CODE STRUCTURE

06 Technical Implementation

The VARSA codebase is structured around a modular, type-safe architecture. Every domain entity (regions, variables, lead times, synoptic regimes, model weights, verification metrics, and extreme hazard thresholds) is codified in strict TypeScript interfaces to eliminate runtime errors.

```
VARSA/ └─ src/ └─ App.tsx # Primary shell, tab routing (5 views), global scenario state └─ index.css #
Tailwind CSS v4 design system imports └─ main.tsx # React 19 root bootstrap └─ components/ └─
ForecastBlendingScreen.tsx # View 2: Model comparison, blend arithmetic, divergence └─
ModelWeightMapScreen.tsx # View 3: Geographic weight distribution & rationale └─ VerificationScreen.tsx #
View 4: Scorecards (MAE, RMSE, Bias, CSI, POD, FAR) └─ ExtremeWeatherScreen.tsx # View 5: Severe hazard
guidance & threshold triggers └─ ForecastChart.tsx # Custom vector SVG multi-line time series visualization
└─ AdaptiveWeightsPanel.tsx # Reusable dynamic weight allocation cards & sliders └─ IndiaMap.tsx #
Regional subdivision geographic reference selector └─ BlendPipelineModal.tsx # Interactive step-by-step
pipeline modal └─ MethodologyModal.tsx # In-app architecture briefing modal └─ data/ └─
varsaData.ts # Met data definitions, subdivisions, verification tables └─ lib/ └─ varsaEngine.ts # Core
computational engine: weights, forecasts, spread └─ package.json # Dependencies & project metadata └─
vite.config.ts # Vite 8 build & bundler configuration └─ vercel.json # Production deployment specification
```

Listing 6.1: VARSA Repository File Hierarchy and Component Responsibilities.

Key Module Responsibilities

Source File	Key Function / Export	Architectural Responsibility
src/lib/varsaEngine.ts	runVarsaDemo()	Single source of truth execution function. Computes adaptive weights, 0–144h time-series curves, spread index, active lead-time point, verification metrics, and hazard alerts.
src/lib/varsaEngine.ts	calculateAdaptiveWeights()	Evaluates region, variable, lead time, and regime rules; enforces strict unit-sum normalization ($w_{GFS} + w_{ECMWF} + w_{ICON} = 1.00$).
src/lib/varsaEngine.ts	categorizeDisagreement()	Computes spread across active members and maps value to Low, Moderate, or High disagreement based on variable-specific thresholds.
src/data/varsaData.ts	REGIONS, VARIABLES	Static meteorological metadata for 6 Indian subdivisions, 3 forecast variables, 5 lead times, and 4 synoptic weather regimes.
src/data/varsaData.ts	VERIFICATION_DATA	Structured demonstration verification scorecards comparing GFS, ECMWF, ICON, Equal Weight, and VARSA across 6 regions and 3 variables.
src/components/ForecastChart.tsx	ForecastChart	Fully custom vector SVG charting component; renders multi-model time series, shaded spread envelopes, observation consensus points, and interactive crosshairs.

SECTION 07 · CORE WEIGHTING FORMULATION

07 Adaptive Weighting Engine

The core innovation demonstrated by VARSA is the transition from static model selection to multi-dimensional context-aware weighting. In `varsaEngine.ts`, the function `calculateAdaptiveWeights()` processes four operational parameters:

- 1. Geographic Sub-Division Specialization:** Sub-divisions with complex topography or coastal interfaces assign greater contribution to models with higher effective resolution or specialized boundary-layer schemes.
- 2. Forecast Variable Physics:** Thermal advection vs. convective precipitation vs. surface wind shear require distinct model combinations.
- 3. Lead Time Degradation Compensation:** At extended forecast horizons (T+72h to T+120h), mesoscale configurations degrade rapidly; the engine automatically reallocates weight toward stable global spectral dynamics (ECMWF).
- 4. Synoptic Regime Modifiers:** Active synoptic features (such as upper-tropospheric Western Disturbances or coastal depressions) apply targeted weight adjustments.

$w_{GFS} + w_{ECMWF} + w_{ICON} \equiv 1.000 \quad (100\%)$

Strict Normalization Guarantee: Raw weight increments are summed ($\Sigma = w_g + w_e + w_p$) and normalized: $w_m = \text{round}(w_m / \Sigma, 2)$. A residual balance term is allocated to the third member to guarantee exact unity without floating-point rounding drift.

Implemented Demonstration Scenarios

To enable transparent, verifiable evaluation during presentations, the repository incorporates four pre-configured demonstration scenarios:

Scenario Preset	Region & Variable	Lead Time & Regime	Allocated Weights	Demonstration Rationale
Scenario A (Vidarbha Basin)	Central India Rainfall (mm/24h)	T+24h Monsoon Convective	GFS 32% ECMWF 46% ICON 22%	ECMWF receives 46% weight for synoptic track representation; GFS contribution reduced to 32%; ICON allocated 22% for convective structure.
Scenario B (Indo-Gangetic Plain)	North India Temperature (°C)	T+48h Pre-Monsoon Heatwave	GFS 28% ECMWF 54% ICON 18%	ECMWF receives 54% weight for synoptic temperature representation; GFS contribution reduced to 28%; ICON allocated 18% for elevation variation.
Scenario C (Peninsular Region)	South India Rainfall (mm/24h)	T+72h Monsoon Convective	GFS 24% ECMWF 52% ICON 24%	Extended range lead time degradation: ECMWF stability increases to 52%; ICON mesoscale weight reduced to 24%; GFS balanced at 24%.
Scenario D (Konkan / Mumbai)	West Coast Wind Speed (km/h)	T+24h Coastal Depression	GFS 17% ECMWF 42% ICON 41%	Maritime boundary shear: ECMWF dominant for offshore synoptic flow (42%); ICON receives 41% for coastal terrain resolution; GFS at 17%.

Table 7.1: Pre-Configured Demonstration Scenarios Implemented in `varsaEngine.ts`.

SCIENTIFIC NOTE ON WEIGHTS

The numerical weights in Table 7.1 represent programmed deterministic logic for prototype demonstration. They are not derived from empirical statistical regression against real historical observations.

SECTION 08 · MATHEMATICAL SYNTHESIS

08 Mathematical Forecast Blending

The VARSA blended forecast is computed via a rigorous, deterministic weighted arithmetic sum across harmonized model member projections.

$$\hat{Y}_{\text{VARSA}}(t) = \sum [Y_m(t) \times w_m(c)] = [Y_{\text{GFS}}(t) \times w_{\text{GFS}}] + [Y_{\text{ECMWF}}(t) \times w_{\text{ECMWF}}] + [Y_{\text{ICON}}(t) \times w_{\text{ICON}}]$$

Where:

- t denotes the valid forecast lead time ($t \in \{0h, 24h, 48h, 72h, 96h, 120h, 144h\}$).
- $Y_m(t)$ is the raw prognostic scalar value for member m (expressed in mm, °C, or km/h).
- $w_m(c)$ is the adaptive weight assigned to model m conditioned on context vector $c = \langle \text{Region, Variable, LeadTime, Regime} \rangle$.
- Constraint: $\sum w_m(c) \equiv 1.000$ strictly across GFS, ECMWF, and ICON.

Step-by-Step Numerical Example (Scenario A: Central India Rainfall at T+24h)

To verify transparent mathematical execution, consider Scenario A as computed directly by `runVarsaDemo()`:

NWP Model Member	Forecast Value (Y_m)	Adaptive Weight (w_m)	Arithmetic Contribution ($Y_m \times w_m$)
NOAA GFS	94.6 mm	0.32 (32%)	$94.6 \times 0.32 = 30.272$ mm
ECMWF IFS	73.1 mm	0.46 (46%)	$73.1 \times 0.46 = 33.626$ mm
DWD ICON	76.8 mm	0.22 (22%)	$76.8 \times 0.22 = 16.896$ mm
VARSA Blended Forecast	Consensus Value	$\Sigma = 1.00$ (100%)	$30.272 + 33.626 + 16.896 = 80.794 \approx 80.8$ mm
Naive Equal-Weight Ensemble	Arithmetic Mean	0.333 each	$(94.6 + 73.1 + 76.8) / 3 = 81.500 \approx 81.5$ mm

Table 8.1: Numerical Demonstration of VARSA Blending Calculation for Scenario A at T+24h.

Mathematical Observations on the Calculation

- In this demonstration scenario, GFS projects an aggressive convective spike (94.6 mm), while ECMWF projects a more moderate accumulation (73.1 mm).
- Because ECMWF receives the highest weight (46%) due to historical synoptic tracking skill, the resulting VARSA blend (80.8 mm) settles closer to ECMWF than naive unweighted averaging would (81.5 mm).
- The calculation is strictly deterministic, rounding the final value to 1 decimal place (`Math.round(val * 10) / 10`).

SECTION 09 · ENSEMBLE SPREAD MECHANICS

09 Model Disagreement & Spread Analysis

Ensemble spread is a foundational concept in operational meteorology. When individual deterministic NWP models converge on the same value, forecaster confidence is elevated. When models diverge substantially, forecast uncertainty increases.

$$\text{Spread}(t) = \max_{m \in M} (Y_m(t)) - \min_{m \in M} (Y_m(t))$$

In `varsaEngine.ts`, Spread is computed as the absolute difference between the highest and lowest member forecasts at the valid lead time.

Implemented Disagreement Thresholds

The function `categorizeDisagreement()` evaluates the computed spread against variable-specific thresholds implemented in the code:

Variable	Low Disagreement (Consensus)	Moderate Disagreement	High Disagreement (High Divergence)
Rainfall	Spread ≤ 8.0 mm	8.0 mm < Spread ≤ 20.0 mm	Spread > 20.0 mm
Temperature	Spread ≤ 1.2 °C	1.2 °C < Spread ≤ 2.5 °C	Spread > 2.5 °C
Wind Speed	Spread ≤ 6.0 km/h	6.0 km/h < Spread ≤ 14.0 km/h	Spread > 14.0 km/h

Table 9.1: Exact Disagreement Categorization Thresholds Implemented in `categorizeDisagreement()`.

Operational Utility as a Contextual Signal

In operational meteorology, high spread does not indicate that all models are incorrect; rather, it alerts the duty forecaster that the atmospheric regime is in a state of high bifurcation sensitivity (e.g., uncertain cyclonic track deflection or convective initiation timing). In the VARSA interface, elevated spread automatically triggers conservative blending adjustments and displays visible alert badges.

Scientific Caveat: In the current prototype, spread serves as a contextual heuristic to alert forecasters to divergence. It should not be interpreted as a calibrated Bayesian credible interval or formal probabilistic ensemble spread, which would require multi-member perturbed-physics ensembles.

SECTION 10 · PHYSICAL PARAMETERS & GUIDANCE

10 Forecast Variables & Output Specifications

The prototype implements complete prognostic curves, tabular matrices, and spatial weighting across three core meteorological parameters critical to Indian disaster management and agricultural operations:

Variable	Physical Unit	Display Precision	Demonstrated Hazard Thresholds
Precipitation / Rainfall	mm / 24h	0.1 mm	Moderate: ≥ 45 mm; Heavy Rain Warning: > 64.5 mm (IMD threshold)
Surface Temperature	°C	0.1 °C	Thermal Advisory: ≥ 41.0 °C; Heatwave Warning: ≥ 42.0 °C
Surface Wind Speed	km/h	0.1 km/h	Squall Watch: ≥ 40.0 km/h; Sustained Gale Warning: ≥ 45.0 km/h

Table 10.1: Forecast Variable Properties and Guidance Thresholds.

Primary Operational Outputs

- **Blended Forecast Point:** The definitive single-value consensus projection at the target lead time.
- **Full Time-Series Vector:** 7 discrete prognostic points (T+0h, T+24h, T+48h, T+72h, T+96h, T+120h, T+144h) enabling trajectory evaluation.
- **Disagreement Spread Index:** Visual uncertainty band showing envelope of model divergence.
- **Extreme Hazard Signals:** Automated detection of threshold exceedances with actionable operational guidance.

Extreme Weather Scenario Guidance Resolution

When blended projections exceed established meteorological hazard triggers, `resolveExtremeGuidance()` generates contextual disaster management advisories. For example, in rainfall scenarios breaching 64.5 mm/24h, the engine generates catchment-level drainage alerts and flags vulnerable sub-districts. In thermal stagnation scenarios breaching 42.0 °C, public health heatwave advisories are automatically formatted.

SECTION 11 · SCORECARD FORMULATION & INTEGRITY

11 Verification Framework

Verification is essential to evaluate forecasting skill. In meteorological literature (Wilks, 2011; Jolliffe & Stephenson, 2012), no single scalar metric captures all dimensions of forecast quality. The VARSA prototype implements a standard six-metric scorecard in VERIFICATION_DATA:

CONTINUOUS VERIFICATION METRICS

1. Mean Absolute Error (MAE):

$MAE = (1/N) \sum |Y_i - O_i|$

Measures average forecast error magnitude without penalizing outliers excessively.

2. Root Mean Squared Error (RMSE):

$RMSE = \sqrt{[(1/N) \sum (Y_i - O_i)^2]}$

Heavily penalizes large forecast errors; sensitive to severe convective misses.

3. Mean Bias:

$Bias = (1/N) \sum (Y_i - O_i)$

Measures systematic tendency to over-predict (>0) or under-predict (<0).

CATEGORICAL CONTINGENCY METRICS (THRESHOLDED)

4. Critical Success Index (CSI / Threat Score):

$CSI = Hits / (Hits + Misses + False\ Alarms)$

Measures accuracy for threshold exceedances (0 to 1, higher is better).

5. Probability of Detection (POD / Hit Rate):

$POD = Hits / (Hits + Misses)$

Proportion of observed extreme events correctly anticipated.

6. False Alarm Ratio (FAR):

$FAR = False\ Alarms / (Hits + False\ Alarms)$

Proportion of forecast event warnings that failed to materialize (lower is better).

CRITICAL VERIFICATION INTEGRITY STATEMENT

The verification values displayed in the current prototype dashboard (such as the illustrative improvement percentage against equal-weight averaging) are simulated demonstration values generated to validate UI scorecard rendering. **They are not evidence of operational forecast-skill improvement.**

Intended Operational Verification Architecture

In a production meteorological deployment, verification metrics must never be computed against in-sample synthetic points. The intended operational evaluation pipeline enforces strict temporal partitioning:

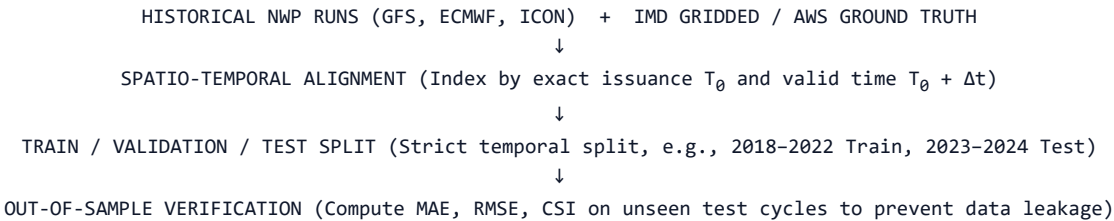


Figure 11.1: Rigorous Operational Verification Flow Preventing Information Leakage.

SECTION 12 · EXECUTION FLOW

12 Current Prototype Workflow

The application executes a 9-step deterministic workflow whenever the forecaster modifies any context control or selects a pre-configured demonstration scenario:

- 1 **Input Ingestion:** Reads active selection for Region (r), Variable (v), Lead Time (l), and Weather Regime (p).
- 2 **Data Harmonization:** Normalizes spatial coordinates to representative station centroid; aligns temporal lead time steps (0h to 144h).
- 3 **Context Analysis:** Evaluates regional terrain elevation, active synoptic jet/depression features, and forecast degradation horizon.
- 4 **Adaptive Weight Determination:** Executes `calculateAdaptiveWeights()`; assigns weights and strictly enforces $\sum w_m = 1.00$.
- 5 **Time-Series & Blend Generation:** Computes raw member curves (Y_{GFS} , Y_{ECMWF} , Y_{ICON}) and computes $\hat{Y}_{VARSA}(t)$ across 7 time points.
- 6 **Disagreement Calculation:** Computes spread at active lead time: $\text{Spread} = \max - \min$; categorizes as Low, Moderate, or High.
- 7 **Verification Lookup:** Retrieves demonstration scorecard metrics from `VERIFICATION_DATA` matching active variable and region.
- 8 **Extreme Hazard Screening:** Compares $\hat{Y}_{VARSA}$ against severe weather thresholds via `resolveExtremeGuidance()`.
- 9 **UI Render:** Dispatches state to React 19 UI tree, updating the active view, SVG chart curves, formula cards, and warning banners.

[Controls State] → [Harmonization] → [Context] → [Weights Engine] → [Blended Series] → [Spread] → [Scorecards] → [UI View]

Figure 12.1: Sequential Execution Pipeline in `runVarsaDemo()`.

SECTION 13 · DASHBOARD & VISUAL DESIGN

13 User Interface & Operational Dashboard

The VARSA frontend provides an interactive, forecaster-centric interface structured into five operational views accessible via the global navigation header:

1. Command Center (src/App.tsx)

The primary operational console. Displays the live scenario selector bar, quick one-click scenario presets (Scenarios A through D), multi-model forecast summary cards, computed VARSA blend, model disagreement gauge, interactive custom SVG time-series chart with model toggle filters, and quick regional weight cards.

2. Forecast Blend (src/components/ForecastBlendingScreen.tsx)

Provides in-depth mathematical visibility into the blending process. Displays individual model forecast contributions side-by-side, explicit arithmetic breakdown of the formula $(GFS \times w_g) + (ECMWF \times w_e) + (ICON \times w_i)$, comparison with the naive equal-weight mean, and a full 7-step lead-time matrix table (T+0h to T+144h).

3. Model Weights (src/components/ModelWeightMapScreen.tsx)

Focuses on geographic and contextual weight distributions. Features an interactive regional selector covering 6 Indian subdivisions, real-time bar breakdowns of model weight percentages, dominant model indicators, and explicit synoptic rationale explaining why specific models receive elevated weight.

4. Verification (src/components/VerificationScreen.tsx)

Displays comprehensive verification scorecards across all three models, equal-weight ensemble, and VARSA. Evaluates MAE, RMSE, Mean Bias, CSI, POD, and FAR in an organized comparison grid, highlighting the relative performance delta across variables and geographic regions.

5. Extreme Guidance (src/components/ExtremeWeatherScreen.tsx)

Specialized disaster management guidance screen. Automatically flags severe weather scenarios when blended consensus values breach established IMD meteorological warning thresholds (e.g., Heavy Rainfall > 64.5 mm, Heatwave ≥ 42.0 °C). Provides operational guidance, severity badges, and affected district sectors.

DASHBOARD INTERFACE LAYOUT SCHEMATIC

```
+-----+
| [NAVBAR] VARSA — Adaptive Weather Intelligence Engine  [A: Central] [B: North] [C: South] [D: West] |
| Tabs: [Command Center] [Forecast Blend] [Model Weights] [Verification] [Extreme Guidance] |
+-----+
| [GLOBAL CONTROLS]: Region: Central | Variable: Rainfall | Lead Time: +24h | Regime: Monsoon |
+-----+
| [METRICS STRIP] GFS: 94.6 mm | ECMWF: 73.1 mm | ICON: 76.8 mm | VARSA BLEND: 80.8 mm (Spread: 21.5) |
+-----+
| [SVG TIME SERIES CHART (0h to 144h)] | [WEIGHT ALLOCATION PANEL] |
| - GFS (Blue) - ECMWF (Amber) - ICON (Teal) | ECMWF: 46% (Dominant) |
| - VARSA Blend (Cyan Line + Points) | GFS: 32% |
| - Spread Envelope (Shaded Grey) | ICON: 22% (Sum = 100%) |
+-----+
| [STEP-BY-STEP ARITHMETIC BREAKDOWN CARD] | [CONTEXTUAL SYNOPTIC RATIONALE CARD] |
+-----+
```

Figure 13.1: Architectural Layout of the VARSA Primary Forecaster Dashboard.

SECTION 14 · SOFTWARE ARCHITECTURE & DEPENDENCIES

14 Technology Stack

The VARSA prototype is implemented using a lean, high-performance web engineering stack. The repository deliberately avoids heavy, black-box charting libraries, implementing all visualizations natively in scalable vector graphics (SVG) to ensure crisp rendering and zero external runtime dependencies.

Layer / Category	Technology	Version	Implementation Role
Frontend Framework	React	19.0.1	Declarative reactive component hierarchy; handles instant state propagation across dashboard views.
Programming Language	TypeScript	7.0.2 / 5.x	Strict static typing across all meteorological entities, weight vectors, and verification scorecards.
Build Tool & Bundler	Vite	8.3.0	Next-generation ESM development server and optimized rollup production bundler.
Styling & Design System	Tailwind CSS	4.3.3	Modern utility-first CSS framework via @tailwindcss/vite plugin; scientific slate/navy visual palette.
Iconography	Lucide React	0.546.0	Crisp, accessible vector iconography for meteorological parameters and system indicators.
Animation & Transitions	Motion (Framer)	12.23.24	Smooth hardware-accelerated transitions between dashboard tabs and status cards.
Data Visualization	Custom Vector SVG	Native	100% custom SVG implementation in ForecastChart.tsx for line charts, shaded spread bands, and axes.
Server / Runtime (Aux)	Express / Node	4.21.2 / 25.x	Auxiliary local API runtime capability for optional backend extensions.
Production Hosting	Vercel Edge Platform	Production	Global CDN edge deployment with automated CI/CD pipeline linked to GitHub repository.

Table 14.1: Complete Implemented Technology Stack (Inspected from package.json).

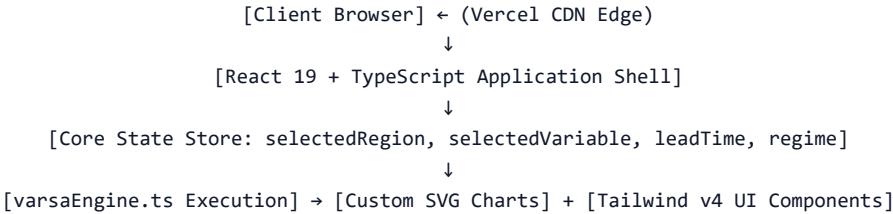


Figure 14.1: Client-Side Runtime Architecture.

SECTION 15 · COMPARATIVE OPERATIONAL ANALYSIS

15 Prototype vs. Intended Operational Architecture

To maintain complete transparency and technical credibility, Table 15.1 explicitly contrasts the current functional prototype against the intended operational enterprise system:

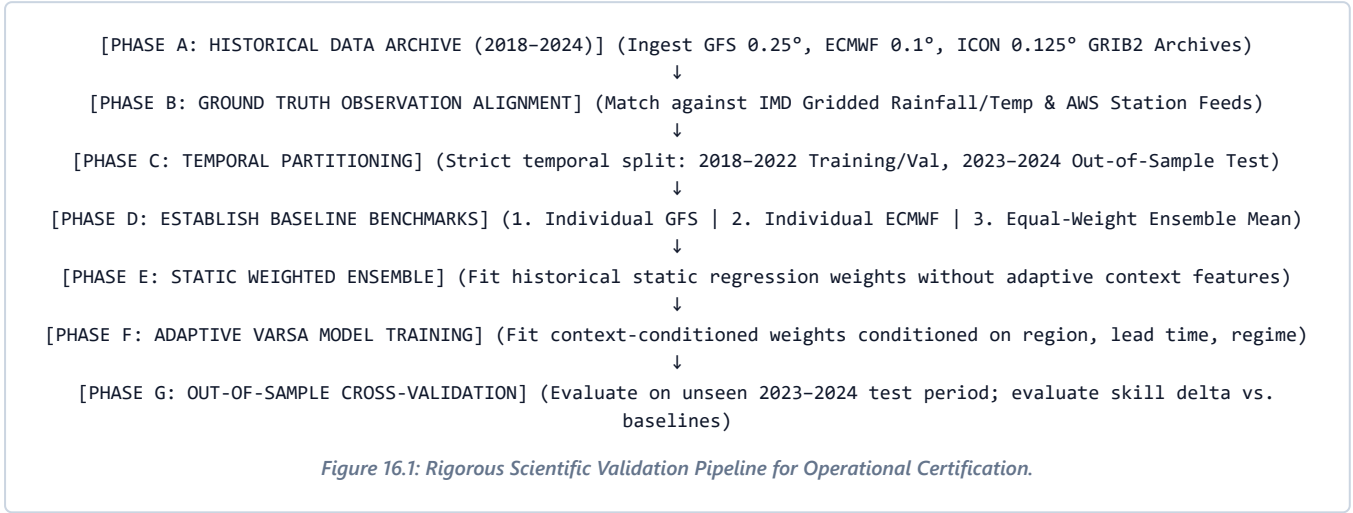
Architectural Dimension	Current Functional Prototype	Intended Operational System
Data Ingestion	Deterministic demonstration dataset representing typical Indian NWP divergence scenarios.	Automated GRIB2/NetCDF ingestion pipelines consuming live NOAA GFS, ECMWF IFS, and NCMRWF feeds via FTP/API.
Observation Ground Truth	Simulated observation reference points designed for interface calibration demonstration.	Automated ingestion of IMD gridded rainfall (0.25°), gridded temperature, Doppler radar mosaics, and AWS telemetry.
Weight Derivation	Heuristic deterministic logic in <code>calculateAdaptiveWeights()</code> conditioned on region, lead time, and regime.	Machine learning models (Gradient Boosted Trees / Regularized Ridge Regression) trained offline on 5+ years of historical verification data.
Spatial Resolution	Sub-division level centroids (6 broad Indian meteorological zones).	High-resolution regularized grid (0.1° × 0.1° / ~10 km over the Indian landmass).
Forecast Computation	Client-side TypeScript execution in the browser within <code>varsaEngine.ts</code> .	Distributed backend pipeline (Python / Ray / Celery) generating gridded spatial netCDF outputs.
Verification Basis	Simulated scorecards demonstrating multi-metric evaluation layout (MAE, RMSE, Bias, CSI).	Rigorous out-of-sample temporal cross-validation preserving strict forecast lead-time causality.
Extreme Guidance	Deterministic threshold triggers demonstrating disaster management advisory workflow.	Calibrated probabilistic thresholding with spatial clustering and catchment-level flood guidance.
Deployment Model	Static Single Page Web Application hosted on Vercel Edge Platform.	Containerized microservices (Docker / Kubernetes) on secure meteorological cloud or on-premise HPC infrastructure.

Table 15.1: Comprehensive Comparison: Current Prototype vs. Intended Operational Architecture.

SECTION 16 · EMPIRICAL VERIFICATION PROTOCOL

16 Scientific Validation Plan

Before any adaptive multi-model blending system can be deployed in operational meteorology, it must undergo rigorous scientific validation across historical seasons. The proposed validation protocol comprises eight sequential phases:



Multi-Dimensional Evaluation Strata

Forecast skill must not be reported as a single nationwide aggregate. Validation must be stratified across:

- **Geographic Sub-Divisions:** Separate verification across all 36 IMD meteorological subdivisions.
- **Seasonality:** Southwest Monsoon (JJAS), Northeast Monsoon (OND), Winter (JF), and Pre-Monsoon (MAM).
- **Lead Times:** Disaggregated tracking at T+24h, T+48h, T+72h, T+96h, T+120h.
- **Synoptic Regimes:** Categorized evaluation during active monsoon depressions, western disturbances, and heatwaves.

Evaluation Beyond Root Mean Squared Error

For precipitation extremes, RMSE is an inadequate metric because it penalizes spatial displacements twice (the "double penalty" problem). The validation protocol mandates:

- **Equitable Threat Score (ETS):** Measures skill in predicting threshold exceedances adjusted for chance hits.
- **Fractions Skill Score (FSS):** Spatial neighborhood verification assessing forecast skill across varying spatial scales (10 km to 100 km).
- **Brier Skill Score (BSS):** Evaluates probabilistic reliability for severe hazard occurrences.

SECTION 17 · TRANSPARENCY & ENGINEERING BOUNDARIES

17 Technical Limitations

In keeping with scientific integrity, the following limitations of the current prototype are documented:

1. **Demonstration Dataset:** Member forecasts and observations in the current repository are deterministically generated to model realistic meteorological behavior; they do not ingest live GRIB2 feeds.
2. **Heuristic Prototype Weights:** Adaptive weights are assigned by deterministic rules in `calculateAdaptiveWeights()`, rather than weights trained on historical observation datasets.
3. **No Live NWP Ingestion:** The repository currently contains no live operational data pipelines to NOAA NOMADS, ECMWF MARS, or IMD data servers.
4. **Absence of Data Assimilation:** VARSA operates purely at the forecast post-processing and blending stage; it performs no physical data assimilation into atmospheric dynamical cores.
5. **No Claim of Validated Operational Skill:** The illustrative skill improvements shown in verification tabs demonstrate UI functionality, not operational superiority over existing IMD forecast systems.
6. **Coarse Spatial Granularity:** The prototype demonstrates six broad regional divisions; it does not resolve 0.1° gridded spatial fields.
7. **Simplified Atmospheric Variables:** The prototype addresses surface rainfall, temperature, and wind; upper-air dynamics (geopotential heights, vorticity, humidity profiles) are not currently modeled.

ENGINEERING CREDIBILITY PERSPECTIVE

Documenting these limitations demonstrates professional engineering rigor. The Arctic Rangers team presents VARSA as a scientifically grounded architecture and interactive proof-of-concept ready for empirical validation, rather than overpromising unverified operational readiness.

SECTION 18 · PHASED DEVELOPMENT PLAN

18 Future Development Roadmap

The operationalization of VARSA into national meteorological centers is planned across seven structured execution phases:

Phase	Target Milestone	Technical Scope & Deliverables
Phase 1	Historical Data Archive	Ingest 5 years (2019–2024) of operational GFS (0.25°), ECMWF IFS (0.1°), and ICON (0.125°) gridded forecasts over the South Asian domain (0°–40°N, 60°–100°E).
Phase 2	Data Harmonization	Build automated regridding pipeline (CDO / xarray) to standardize all models onto a common 0.1° regular grid; align temporal timesteps.
Phase 3	Ground Truth Alignment	Collocate harmonized model grids with IMD daily gridded rainfall (0.25°) and temperature (0.5°) datasets, plus automatic weather station telemetry.
Phase 4	Adaptive ML Model	Train gradient boosted decision trees (XGBoost / LightGBM) to predict optimal model weights conditioned on geographic, temporal, and synoptic feature vectors.
Phase 5	Out-of-Sample Validation	Execute out-of-sample temporal testing on 2023–2024 seasons; benchmark against individual models and unweighted ensemble averages using ETS, FSS, and RMSE.
Phase 6	Operational Pipeline	Deploy automated backend microservices to fetch daily operational 00Z/12Z model cycles, execute inference, and generate gridded NetCDF outputs.
Phase 7	Production Deployment	Integrate into operational forecaster workstations; deliver API endpoints for disaster management authorities and regional meteorological centers.

Table 18.1: Phased Development Roadmap Toward Operational Deployment.

SECTION 19 · TECHNICAL SUMMARY

19 Conclusion

VARSA establishes a viable architectural and mathematical framework for hybrid multi-model forecast blending across the complex meteorological terrain of India. By recognizing that no individual numerical weather model is universally superior across all regions, seasons, variables, and lead times, VARSA demonstrates how dynamic, context-aware weighting can systematically synthesize complementary model strengths while mitigating systematic biases.

The current repository delivers a complete, interactive, and mathematically sound functional prototype. It proves the feasibility of dynamic weight allocation ($\sum w_m = 1.000$), transparent step-by-step arithmetic blending, spread-based disagreement classification, multi-metric verification scorecards, and automated extreme hazard thresholding. The next milestone requires transition to Phase 1 through Phase 5 of the scientific roadmap, training adaptive blending models over extensive historical NWP archives paired with IMD observational ground truth.

SECTION 20 · SCIENTIFIC CITATIONS

20 References & Scientific Literature

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